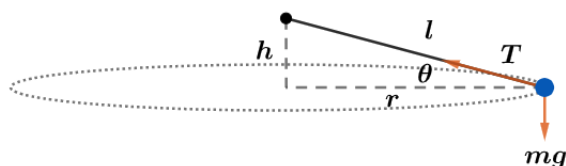


2023 F=ma Exam: Problem 6

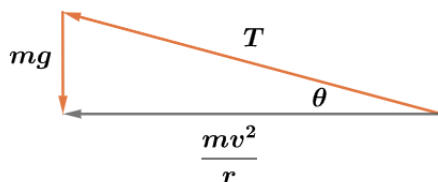
Kevin S. Huang



We have

$$\sin \theta = \frac{h}{l}$$

where h is the height difference between the ends of the rope.



Considering the force vectors, we have

$$\tan \theta = \frac{mg}{mv^2/r} = \frac{gr}{v^2}$$

Since $h \ll l$, we can make the small angle approximation $\tan \theta \approx \sin \theta$,

$$\frac{gr}{v^2} = \frac{h}{l}$$

Furthermore, $\cos \theta \approx 1$ so $r = l \cos \theta \approx l$. Substituting this in the previous equation,

$$\frac{gl}{v^2} = \frac{h}{l}$$

$$h = \frac{gl^2}{v^2} = \frac{(10 \text{ m/s}^2)(0.5 \text{ m})^2}{(15 \text{ m/s})^2} = 1.1 \text{ cm}$$

so the answer is A.